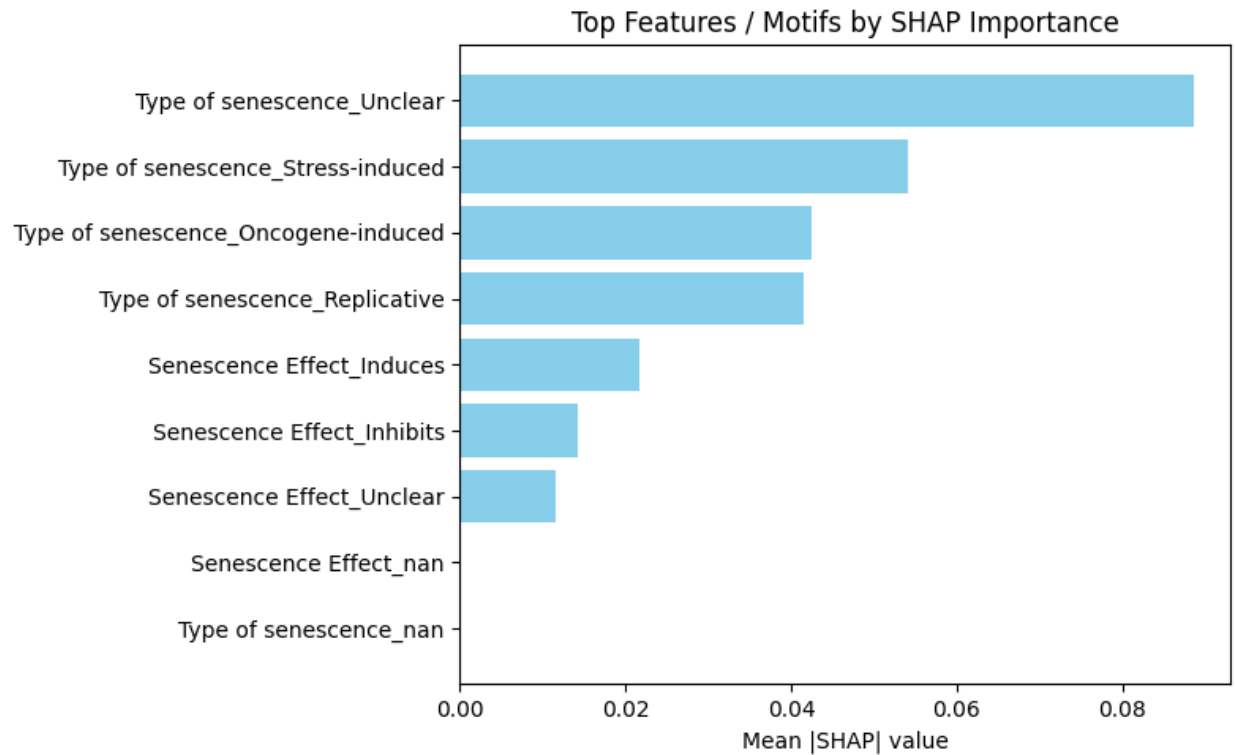
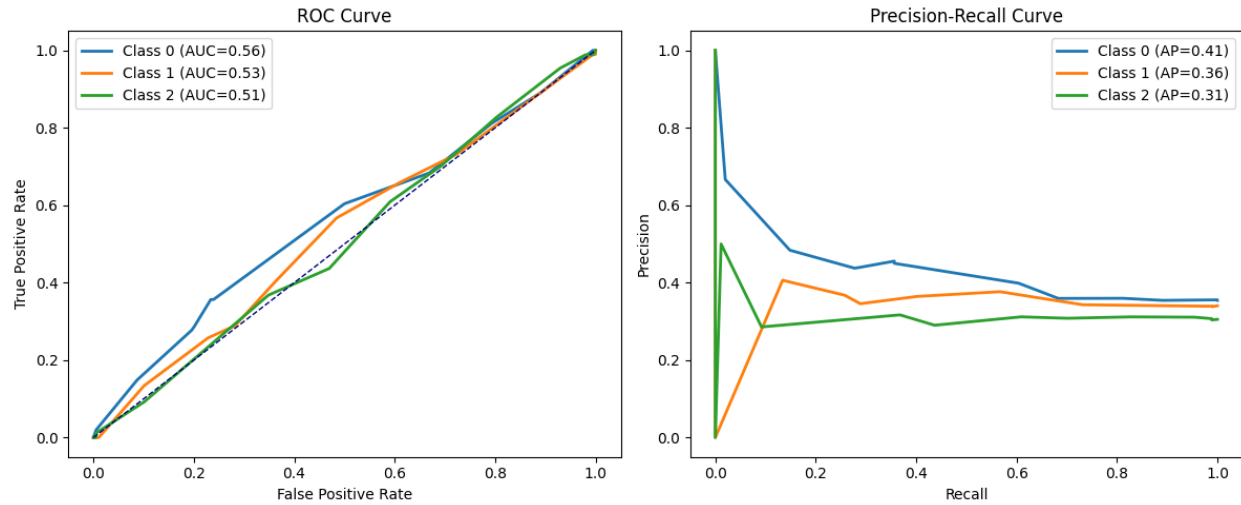




Caption 1 (ROC/PR): ROC/PR curves show how well the model distinguishes between classes; class 0 is shown as an example.

Caption 2 (Motif importance): Top 10 features/motifs influencing the model predictions according to SHAP values.

Methods:

1. Dataset Preparation: Collected gene and promoter datasets from multiple sources. Generated a labeled slice (approximately 5–10k cells) to facilitate rapid iteration. Cleaned, encoded, and split features and labels for model training.
2. Baseline Model Training: Trained three state classifiers – Logistic Regression, Gradient Boosting (GBM), and a small neural network – to predict senescence status of genes. Evaluated using AUROC, AUPR, and confusion matrices.
3. Motif Scan: Conducted enrichment analysis on top and anti-senescent genes to identify candidate regulatory motifs. Selected top motifs based on occurrence and significance.
4. Safety Assessment: Defined leakiness and off-target test cases. Computed safety scores for each promoter, producing a “safe / not safe” rubric for candidate circuits.
5. Circuit Design & Simulation: Constructed basic genetic logic circuits (AND, NOT, OR). Simulated circuit behavior to verify ON/OFF states and ablation effects.
6. Explainability: Applied SHAP analysis on GBM predictions to identify top features/motifs influencing model output. Generated ROC and Precision-Recall curves and feature importance charts with captions.

Results:

- Logistic Regression, GBM, and small NN achieved moderate performance (AUROC approximately 0.60–0.62 and AUPR approximately 0.43–0.45), indicating reasonable predictive power for senescence classification.
- Motif enrichment highlighted several candidate regulatory motifs (TTTTTT, AAAAAA) associated with top and anti-senescent genes.
- Safety analysis revealed differential promoter leakiness and off-target effects, enabling a first-pass categorization of circuits as “safe” or “not safe.”
- Circuit simulations confirmed expected logic behavior (AND, NOT, OR) and demonstrated changes upon single-component ablation.
- SHAP analysis indicated “Type of senescence” and “Senescence Effect” as the most influential features in model predictions, providing interpretability for baseline models.

Limits:

- The predictive models are trained on a limited labeled subset; generalization to all genes or cell types is not guaranteed.
- Safety scores are first-pass estimates based on in silico assessments; experimental validation is required.
- Circuit simulations are simplified and do not account for full cellular context or complex regulatory interactions.
- Motif enrichment identifies associations, not causal regulatory mechanisms.
- The current study does not provide clinical or therapeutic recommendations; findings are exploratory and computational.